

Mini Hybrid V2.2 · 2026-09-21 · generated from configs/V2.2_recommended_md01.0_HEM.json

The short answer

Specific impulse rises monotonically across the whole range you asked about: 195.0 s at 1.0 kg/s to 203.4 s at 1.5 kg/s, a gain of 4.3%. There is no interior optimum to find. But that gain is bought almost entirely with chamber pressure, and every constraint you have runs the other way.

Take 1.0 kg/s. The extra 4% of Isp at 1.5 kg/s costs you 50% more oxidiser flux in a grain that is already above the correlation range, a stability margin that expires at 60% of the burn instead of 84%, and 7 percentage points of unburnt fuel. None of those are modelled risks you can buy back later.

Why Isp rises — and why that is not really about O/F

Your nozzle ($\epsilon = 4.340$) is perfectly expanded at $P_c = 24.9$ bar. At 1.0 kg/s the chamber peaks at 25.3 bar, so the motor spends most of the burn overexpanded; pushing to 1.5 kg/s raises the peak to 37.5 bar and the thrust coefficient with it, from 1.410 at 20 bar to 1.509 at 37 bar. That is a +7% effect.

O/F moves the opposite way, from 8.13 to 9.07, and the CEA table puts peak Isp at O/F 7.00. So raising the flow walks you away from the mixture optimum — it is worth about -1.4%, which the nozzle gain simply outweighs. The Isp trend is a nozzle-matching result wearing a mixture-ratio costume.

Worth knowing: the propellant file declares $opt_OF = 8.27$, but its own CEA table peaks at O/F 7.00. The declared value is an HRAP default and does not match the table it ships with. Nothing in the sizing uses it except as a fallback, but do not design to it.

What actually limits you

- Oxidiser flux. G_{ox} starts at 797 kg/m²/s at 1.0 kg/s and 1205 at 1.5. Paraffin correlations are normally fitted below ~700. You are extrapolating the regression law at both ends of the range, and G_{ox} is set by the grain port — no injector fixes it.
- Feed isolation. The 20% pressure-drop margin lasts 84% of the liquid phase at 1.0 kg/s but only 60% at 1.5 kg/s. The late burn is where feed coupling bites.
- Fuel utilisation. 75% of the grain is consumed at 1.0 kg/s, 67% at 1.5. Faster burns leave more paraffin behind.
- Chamber pressure. 1.5 kg/s puts 37.5 bar into a case you sized around a much lower number. That is a structural question, not a performance one.

The recommendation

41 holes	Ø1.20 mm	THRU	plate	85.9 mm × 12.7 mm	thick
L/D	10.58	→	HEM regime	CdA	32.46 mm ² min web 9.55 mm
$\dot{m}_{ox}$	1.0 kg/s	O/F	8.13	P_c	25.3 bar peak / 21.2 mean
thrust	1885 N	mean	burn	6.89 s	total impulse 9468 N·s Isp 195.0 s

Hole diameter is held at 1.20 mm because $L/D = 10.6$ is the only setting on a 12.7 mm plate that lands inside the L/D band where the Waxman cold-flow data validates HEM. Vary the hole count, not the diameter — that keeps the flow model valid as the flow target moves.

The variable, and the thing that must not vary with it

Hole diameter is held at 1.20 mm and hole count is solved for. That is the opposite of the usual habit, and it is deliberate: L/D = thickness / diameter, so holding the diameter holds L/D at 10.58 across every point in the study. Had the diameter been solved for instead, L/D would have drifted from 12.7 to 6.4 across the sweep, silently crossing out of the regime where HEM applies and into the one where Dyer does — and the endpoints would no longer be comparable, because they would be different physics rather than different sizes.

Run settings

chamber model	quasi-steady	(timestep-independent; see § convergence)
timestep	1 ms	
sizing objective	mdot_ox	(never inverts the regression law)
flow model	HEM	(L/D 10.58; externally validated band)
discharge coeff	0.70	(placeholder – not yet measured)
property backend	ESDU 91022	(HRAP MATLAB parity)
grain	5 ports × 17.78 mm ID, OD 85.85 mm, L 178 mm	
tank	7.0 L, 4.480 kg N ₂ O, 23.9 °C saturated	
nozzle	throat 30.48 mm, ϵ 4.340, Cd 0.95, η 0.9	
propellant	a 0.0304, n 0.681 -- HRAP defaults, NOT measured	

Sweep results

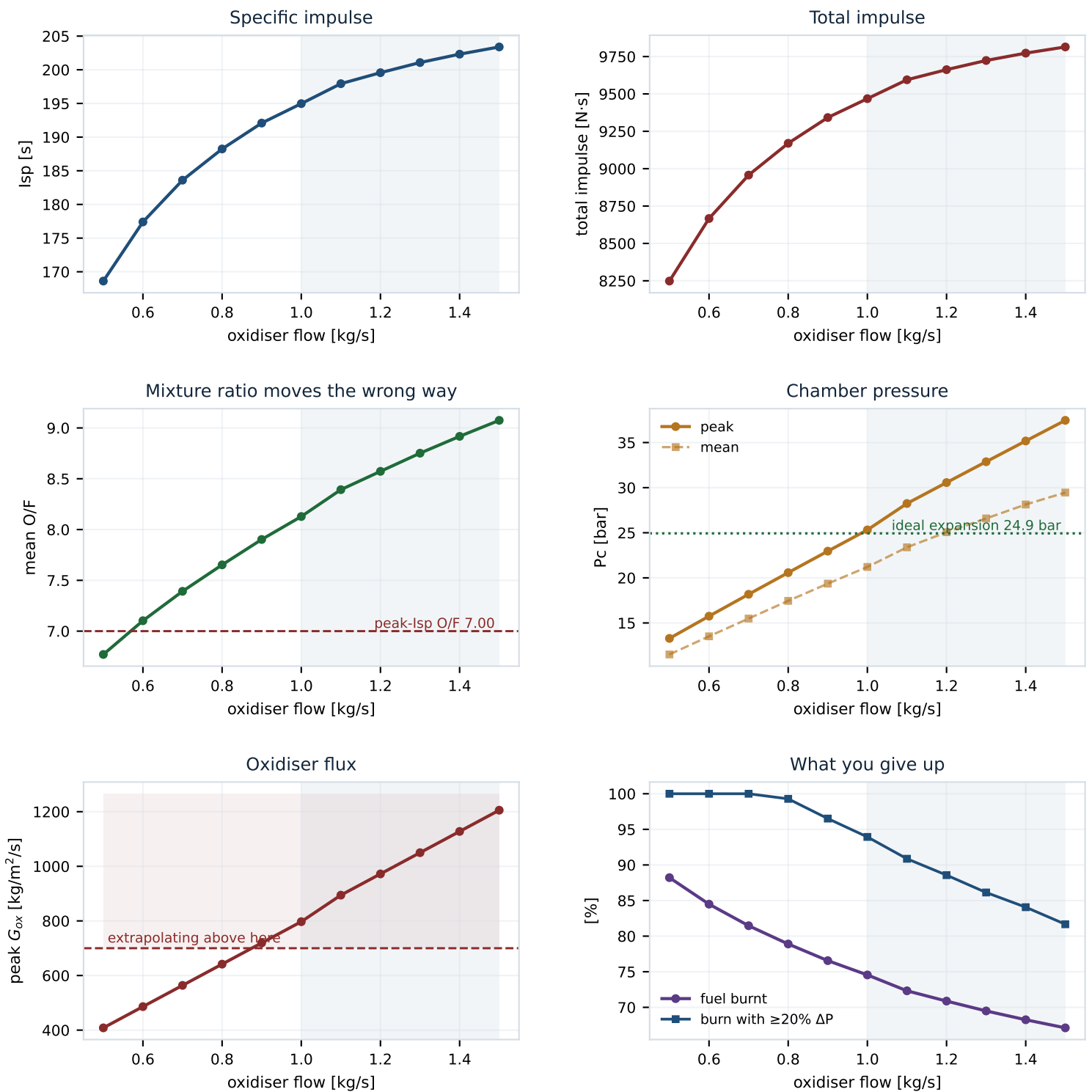
Isp is over the whole run including the vapour tail. Gox>hi% is the share of the liquid phase spent above the fitted flux range; dPok% is the share with at least 20% pressure-drop margin; fuel% is the grain actually consumed.

mdot	N	CdA	Pc pk	Pc mn	F mean	burn	Itot	O/F	Isp	Gox	Gox>hi%	fuel%	dPok%	OF>10%	web
0.50	21	16.63	13.3	11.5	896	12.04	8248	6.77	168.6	408	0	88.2	100	0	14.93
0.60	25	19.79	15.8	13.5	1101	10.40	8667	7.10	177.4	486	0	84.5	100	0	13.03
0.70	29	22.96	18.2	15.5	1302	9.20	8957	7.39	183.6	564	0	81.5	100	0	11.04
0.80	33	26.13	20.6	17.5	1502	8.28	9170	7.65	188.2	642	0	78.9	99	0	9.27
0.90	37	29.29	23.0	19.4	1697	7.50	9342	7.90	192.1	719	1	76.6	97	0	8.42
1.00	41	32.46	25.3	21.2	1885	6.89	9468	8.13	195.0	797	4	74.6	94	0	9.55
1.10	46	36.42	28.2	23.4	2107	6.26	9595	8.39	197.9	894	9	72.3	91	0	8.28
1.20	50	39.58	30.6	25.1	2277	5.89	9662	8.57	199.6	972	13	70.9	89	0	7.91
1.30	54	42.75	32.9	26.6	2432	5.54	9724	8.75	201.1	1050	17	69.5	86	0	7.28
1.40	58	45.92	35.2	28.1	2590	5.26	9773	8.92	202.3	1128	21	68.3	84	0	6.46
1.50	62	49.08	37.5	29.5	2725	4.99	9814	9.07	203.4	1205	24	67.1	82	13	6.19

Reading it

- Isp and total impulse both increase all the way to 1.5 kg/s. If performance were the only axis, the answer would be 'as much flow as the plate can pass'.
- Every other column degrades monotonically in the same direction. This is a constraint problem, not an optimisation problem — there is no peak to find, only a line to stop at.
- The range below 1.0 kg/s is included because the O/F optimum lies below the brief's range. An optimum you cannot see both sides of is not an optimum.

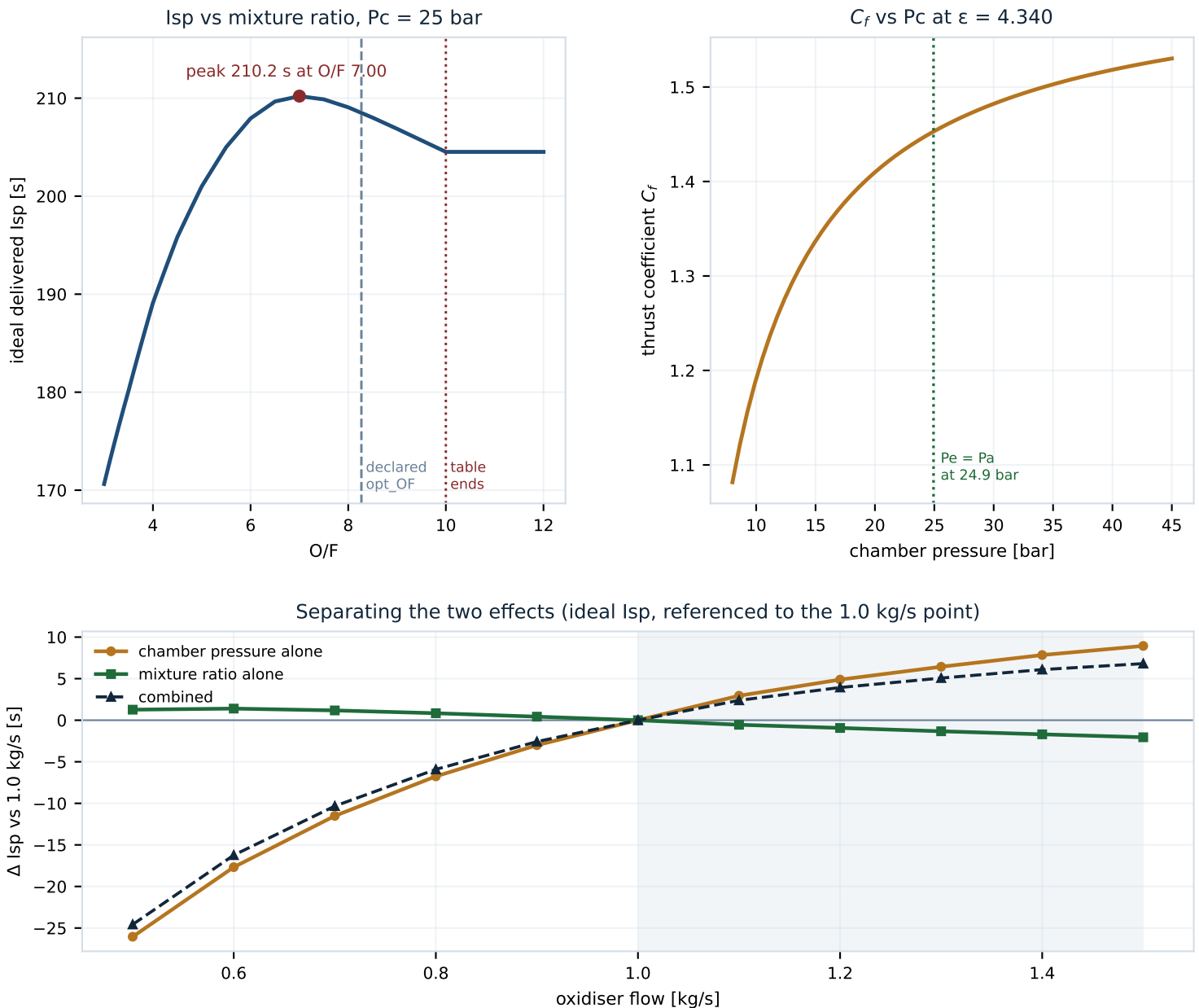
OF>10% is the share of the run whose instantaneous O/F is past the end of the CEA table, where c^ is clamped rather than computed. It is zero up to 1.4 kg/s and 13% at 1.5 kg/s — so the top row of this table is the one row partly resting on an extrapolated c^* , and it is also the row being argued against.*



The shaded band is the range in the brief. Read the top row against the bottom row: everything that improves with flow is on the top, everything that degrades is on the bottom, and they cross nowhere. The choice is therefore a judgement about which constraint you are least willing to violate, not a maximum.

Where the Isp gain actually comes from

Between 1.0 and 1.5 kg/s the thrust coefficient rises 7.1% (nozzle matching, 1.410→1.509) while c^* falls about 1% (mixture ratio drifting past its optimum). The net is the +4% you see. If the nozzle were re-cut for the lower chamber pressure, most of the apparent advantage of running hard would disappear.



Two things the left-hand plot says

- Peak Isp is at O/F 7.00, not at the 8.27 the propellant file declares. The declared figure is an HRAP default that does not agree with the CEA table shipped alongside it. Design to the table.
- The curve is flat near the top: O/F 6.0 to 8.5 costs under 1% of Isp. Mixture ratio is simply not where your performance is being won or lost — which is fortunate, because it is also the number you know least well.
- Above O/F 10 the CEA table ends and c^* is clamped to its last value. Any run that drifts past 10 is reporting a flat extrapolation, not a computed c^* . At a $\times 0.7$ the sweep reaches O/F 12.6 and those Isp figures are not meaningful.

The right-hand plot, and the decomposition below it

C_f climbs steadily with chamber pressure and keeps climbing past the 24.9 bar ideal-expansion point, because for a fixed area ratio the pressure-thrust term keeps growing. The bottom plot holds one variable at its 1.0 kg/s value and moves the other, so the two contributions can be read apart rather than only as their sum. Going to 1.5 kg/s, chamber pressure is worth +8.9 s and the mixture-ratio drift -2.1 s.

That is the answer to 'optimise O/F and Isp' in one line: in this motor they are not the same optimisation, and they are not the same size. Over the brief's range chamber pressure is worth +8.9 s and mixture ratio -2.1 s — a factor of four apart, and over the full sweep a factor of ten. Chamber pressure here is a consequence of a nozzle that is large for the flow, not of anything the injector is doing. Optimising the mixture ratio is polishing the smaller term.

1. Oxidiser flux — the binding one

$G_{ox} = \dot{m}_{ox} / A_{port}$, and A_{port} is 1241 mm² at ignition. That fixes the flux entirely from the grain: the injector has no say in it. Across the brief's range the starting flux runs from 797 to 1205 kg/m²/s, against a correlation range normally fitted below ~700.

$\dot{m}_{dot}$	peak G_{ox}	% of burn above fit range	port ID for $G_{ox} < 700$
1.00	797	4	19.1
1.10	894	9	20.0
1.20	972	13	20.9
1.30	1050	17	21.7
1.40	1128	21	22.6
1.50	1205	24	23.4

Flux falls quickly as the port opens, so only the early burn is affected — but the early burn is where most of the fuel flow is. Opening the ports from 17.78 mm to about 19 mm would bring 1.0 kg/s inside the fitted range; 1.5 kg/s would need 23.4 mm, which the 85.85 mm grain cannot give you with five ports and any useful web left.

2. Feed isolation

A bigger injector passes more flow, which raises chamber pressure, which eats the pressure drop. Late in blowdown the tank has decayed and the margin goes with it. The number that matters is not the minimum — that always collapses at the very end — but how much of the burn is spent above 20%.

$\dot{m}_{dot}$	margin expires at	liquid phase	clean fraction
0.80	8.13 s	8.28 s	98%
1.00	5.80 s	6.89 s	84%
1.20	4.26 s	5.89 s	72%
1.50	2.98 s	4.99 s	60%

3. Manufacturability — not a constraint here

Every hole count in the sweep is comfortably drillable. Even 62 holes of 1.20 mm leaves a 6.19 mm web and a 5.15 mm edge margin on an 85.85 mm plate. The plate is the one part of this design with margin to spare.

4. Timestep — checked, not assumed

dt [ms]	holes	impulse [N.s]	I_{sp} [s]
4.00	41	9472	195.10
2.00	41	9476	195.16
1.00	41	9468	194.97
0.50	41	9464	194.85
0.25	41	9474	195.09

Impulse varies by 0.1% over a 16× change in timestep, and the hole count does not move at all. The quasi-steady chamber model has no timestep sensitivity by construction; this confirms the tank blowdown integration does not either at these settings.

Three inputs here are genuinely uncertain: the flow model, the discharge coefficient, and the regression coefficients. The question is not whether they change the numbers — they do — but whether they change the decision.

Flow model — changes the plate, not the conclusion

	1.00 kg/s			1.25 kg/s			1.50 kg/s		
SPI	19 holes	190.1s		26 holes	195.5s		37 holes	200.5s	
Dyer	26 holes	192.0s		35 holes	198.3s		46 holes	201.9s	
HEM	41 holes	195.0s		52 holes	200.3s		62 holes	203.4s	

The hole count varies by more than $2\times$ between SPI and HEM at the same flow, because the models disagree by that much about how much area a given mass flow needs. The lsp ranking is untouched: every model says more flow is better, and they agree on the value to within 5 s. The model uncertainty lands entirely on the hardware and not at all on the trade.

Discharge coefficient — same story, smaller

	1.00 kg/s			1.25 kg/s			1.50 kg/s		
Cd 0.60	48 holes	195.1s		60 holes	200.2s		73 holes	203.5s	
Cd 0.70	41 holes	195.0s		52 holes	200.3s		62 holes	203.4s	
Cd 0.80	36 holes	195.1s		45 holes	200.2s		54 holes	203.3s	

A Cd error is a pure area scaling: $\pm 14\%$ in Cd moves the hole count by $\pm 17\%$ and the lsp by 0.1 s. Cd matters enormously for building the right plate and not at all for choosing the operating point.

Regression coefficients — the one that hurts

	1.00 kg/s			1.25 kg/s			1.50 kg/s		
a x 0.70	O/F	11.27	190.1s	O/F	12.04	197.2s	O/F	12.61	201.3s
a x 1.00	O/F	8.13	195.0s	O/F	8.66	200.3s	O/F	9.07	203.4s
a x 1.30	O/F	6.41	197.2s	O/F	6.82	203.9s	O/F	7.14	207.5s
a x 1.60	O/F	5.50	201.9s	O/F	5.81	207.2s	O/F	6.00	208.8s
a x 2.30	O/F	3.92	187.1s	O/F	4.23	196.7s	O/F	4.46	202.7s

Scaling a from $0.7\times$ to $2.3\times$ — which is roughly the span between the HRAP default and what your own HRAP run implied — moves the mean O/F from 12.6 to 4.5. That is the whole usable mixture range. lsp moves far less, between 187 and 209 s, because the lsp-vs-O/F curve is flat near its peak.

Crucially the ranking holds in all five cases: at every value of a, more oxidiser flow gives more lsp. So the decision is robust even though the predicted O/F is not. That is the reason to size on oxidiser flow and then find out what O/F you got, rather than the other way round.

At a x 0.7 the mean O/F reaches 12.6, past the end of the CEA table, so those lsp figures rest on a clamped c. Treat that row as directional only.*

Open the grain ports

This is the highest-value change available, and it is the only one that addresses the binding constraint. Going from 17.78 mm to 19.1 mm ports brings 1.0 kg/s inside the fitted flux range. It also lowers O/F, because a larger port at the same oxidiser flow means lower flux, slower regression — and here the effect runs the useful way, toward the flat top of the Isp curve. If the grain is not yet cast, do this.

Supercharge the tank

feed at 1.50 kg/s	holes	Isp	Pc peak	clean margin	peak Gox
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saturated 23.9 C	62	203.4s	37.5 bar	82%	1205
saturated 30 C	58	202.7s	37.3 bar	82%	1201
supercharged 65 bar	48	215.4s	51.8 bar	100%	1214
supercharged 75 bar	35	214.4s	51.0 bar	100%	1220

Supercharging to 65 bar produces the best result anywhere in this study: 215.4 s at 1.5 kg/s with the 20% margin intact for the entire burn. The mechanism is simple — subcooling the feed raises the driving pressure, so the same flow needs less area, and chamber pressure climbs to 51.8 bar where the nozzle is far better matched.

Two cautions. This model holds the supercharge pressure constant for the whole burn; a real blowdown-pressurised tank decays, so the margin shown is optimistic unless the supply is regulated. And 51.8 bar chamber with 65 bar feed is a different pressure vessel problem than the one you have costed. Supercharging does nothing at all for G_{ox}.

Warm the fill

Filling at 30 °C instead of 23.9 °C is nearly free and does very little: Isp 202.7 vs 203.4 s. Higher saturation pressure means more flow per unit area, so the solver just gives you fewer holes. Not a lever worth pulling for performance, though it does help repeatability if you can control it.

Re-cut the nozzle

Not in scope here, but worth stating plainly since it drives the whole result: $\epsilon = 4.340$ is matched to 24.9 bar, and at 1.0 kg/s you run below that for most of the burn. A smaller throat would raise chamber pressure at the same flow and capture most of the benefit that this study attributes to running harder — without the flux penalty. That is the trade worth running next.

Every number here is conditional on the inputs below. They are listed in descending order of how much they could move the answer.

Unmeasured inputs

- Regression coefficients a and n are HRAP defaults, and HRAP ships the same pair for ABS, asphalt and HTPB — they are placeholders, not paraffin properties. Your own HRAP run implied a value about 2.3× larger. Nothing here resolves that; only a hot fire does.
- $C_d = 0.70$ is assumed. A water cold-flow test measures it directly and is the single highest-value test available before firing.
- The flow model is chosen from L/D, not measured. Only a cold flow with a flashing fluid distinguishes HEM from Dyer at this L/D.
- c^* efficiency is 1.0 and nozzle efficiency 0.9. Real injector mixing efficiency on a showerhead plate is typically below unity, so delivered I_{sp} will be under these figures.

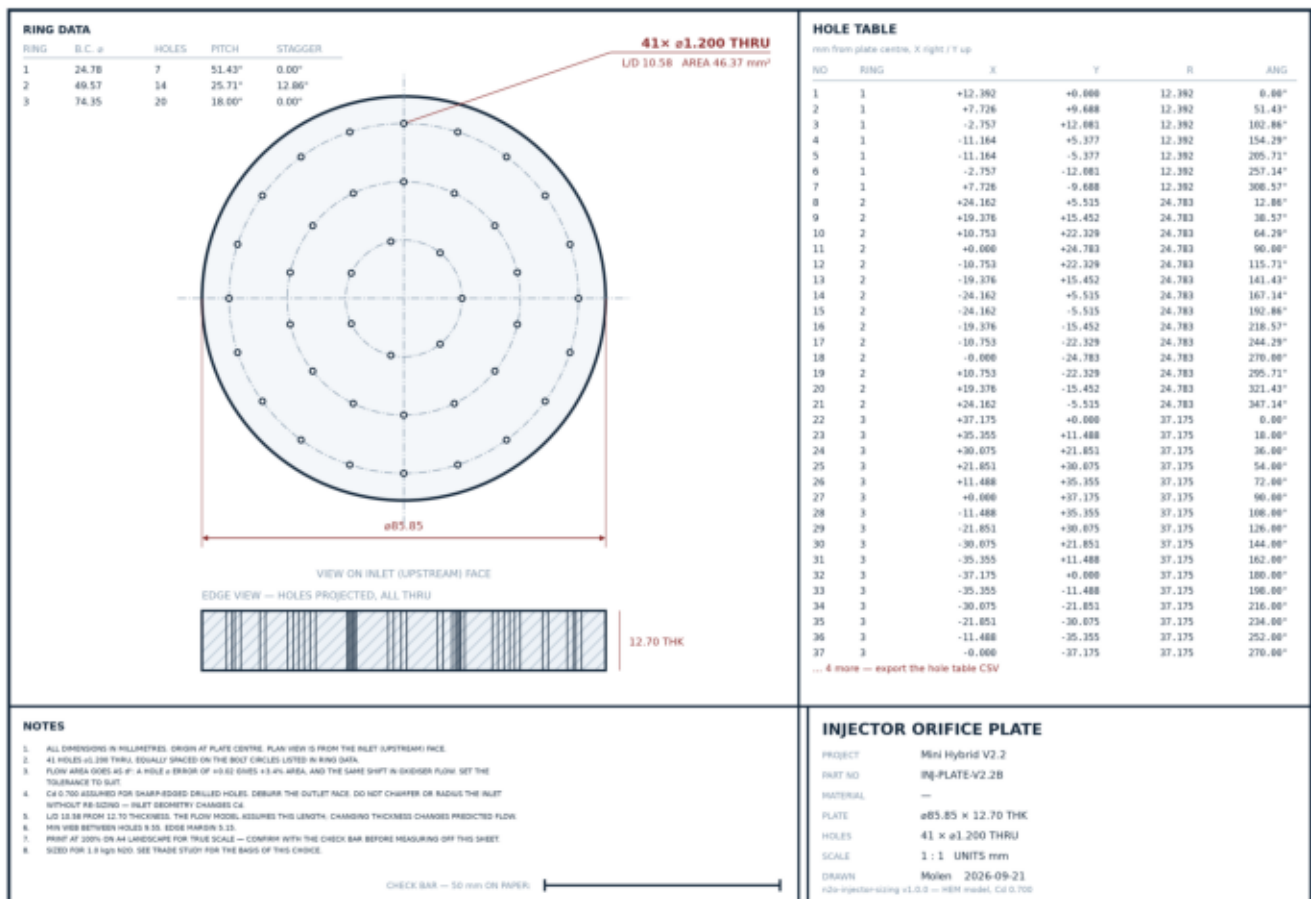
Model limitations that matter at this operating point

- No combustion instability model. The 20% pressure-drop criterion is a rule of thumb, not a prediction. Chugging is not simulated.
- No throat erosion. Over a 7 s burn with a graphite throat this is small but not zero, and it would reduce chamber pressure late in the burn.
- Tank blowdown is adiabatic with liquid and vapour in equilibrium. Real tanks draw heat from the walls, so pressure decays more slowly and the real burn is likely a little longer than predicted.
- The supercharged case holds feed pressure constant, which no unregulated system does.
- The grain is modelled as five identical circular ports with no end-face regression and no port-to-port coupling.

How to close the gaps, cheapest first

1. water cold flow of the plate -> C_d , and confirms the drilling
2. N2O cold flow -> settles SPI / Dyer / HEM
3. one hot fire, fuel weighed before/after -> a and n for your paraffin
4. second fire at a different flow -> separates a from n

Until step 3, treat the O/F column of this report as an order-of-magnitude statement and the I_{sp} column as a comparison between options rather than an absolute prediction. The trade conclusion — that 1.0 kg/s is the right end of the range — does not depend on any of it, which is the one genuinely firm result here.



Full-size sheets, the DXF and the hole coordinate table are produced from the Plate drawing tab, or from the configuration file saved alongside this report. The drawing and the DXF are generated from the same layout object, so they cannot disagree about where a hole is.

Before this is cut

- Cold-flow the plate in water to measure Cd. At the assumed 0.70 the hole count is 41; at a measured 0.60 it would be 48 and at 0.80 it would be 36. This is the one number that changes the part.
- Decide the hole-diameter tolerance. Flow area goes as d^2 , so +0.02 mm on ϕ 1.20 is +3.4% area and the same again in oxidiser flow.
- Confirm the plate outside diameter and the seal arrangement against the case. The 85.85 mm here is the grain outer diameter, used as a placeholder envelope.

In order

- size on oxidiser flow, not O/F
 - open the grain ports to ~19 mm
 - water cold flow the plate
 - make two plates, not one
 - first fire, weigh the fuel
 - then revisit the nozzle
- already done -- 1.0 kg/s, 41 x 1.20 mm
if the grain is not yet cast
measures Cd; re-run with the real number
the model spread is 2.2x in area
gives a and n; everything else follows
the largest single lever left

Steps 1 and 2 are free. Step 3 costs an afternoon and removes the largest avoidable uncertainty in the design. Nothing before step 5 can tell you what O/F this motor will actually run at.